Supplementary Material:

Every Fact, and how it was obtained

for *The Internal Space and Boundary Growth of the Spectre and Tsiokos's Chair44 Tilings*

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What this is

The main paper stratifies its statements: *Facts* are exact statements, each proved, verified by an exact finite computation over a stated range, or quoted; *Propositions* carry a modelling step. This document lists every Fact of the paper — all 71 — with the key under which the ledger (`spectrefacts.py`, `chair44facts.py`, `monotilefacts.py`) computes it, the claim, the first sentence of the method by which it was obtained, and a mark on the 24 Facts whose ledger entry names a theorem of the Lean 4 file. The ledger is shared with other documents; their Facts are not listed here.

Three things about the table.

- **It is generated, not written.** Every row is emitted at build time from `spectrefacts.py`, the same module that produces every number in the main paper. There is no second copy to drift.
- **The Lean column is measured, not asserted.** A checkmark means the Fact's `lean` field names a theorem that exists in `SpectreFacts.lean` — 67 theorems, kernel-accepted, no admitted proofs — and the paper's build fails if a named theorem is missing; six phantom attributions were found and removed when that check was added. A Lean theorem proves the Lean statement, which can be narrower than the paper's: the Facts marked * are proved in Lean for the sequence that the Lean file *defines* by the perimeter recurrence. That this sequence is the supertile perimeter at every order is Theorem 1 of the paper, proved by an exact finite certificate (`boundary_certificate/`), not in Lean.
- **The method column is the first sentence.** A method that runs to a paragraph is truncated at its first full stop or semicolon. The full text is in the module, one lookup by key away.

Two of the Facts are inputs rather than results, and the paper says so: `charpoly` rests on the transcribed substitution matrix, and `boundary_recurrence` on perimeters counted from placed tiles for $k \leq 6$; beyond $k = 6$ the recurrence is a conjecture. Both are audited by an independent route — the matrix against a second implementation of the same substitution, the seed against an exact $\mathbb{Q}(\sqrt{3})$ recount — and an audit is not a proof. They are the two rows a sceptical reader should look at first.

The table

Generated from `spectrefacts.py`. The last column marks the Facts that are also proved in Lean, see https://github.com/brentharts/spectre/wiki/Spectre-Lean; the others are computations.

Key	Claim	How	Lean
`charpoly`	$\chi_M(x) = x^5(x-1)(x+1)(x^2-8x+1)$	exact characteristic polynomial over Z, factored	
`perron`	$\lambda^2 = 4 + \sqrt{15}$	largest root of the quadratic factor	
`pell`	$4^2 - 15 \cdot 1^2 = 1$	the Pell equation, so lambdaˆ2 is the fundamental unit of Z[sqrt15]	✓
`lam_closed`	$\lambda = \frac{1}{2}(\sqrt{6} + \sqrt{10})$	square both sides: (sqrt6+sqrt10)ˆ2/4 = (16+4 sqrt15)/4	
`minpoly`	$x^4 - 8x^2 + 1$	minimal polynomial of lambda over Q	
`channel_hex`	$(\lambda - \lambda^{-1})^2 = 6$	lambdaˆ2 + lambdaˆ-2 - 2, in Z[sqrt15], no root evaluated	✓
`channel_pent`	$(\lambda + \lambda^{-1})^2 = 10$	lambdaˆ2 + lambdaˆ-2 + 2, in Z[sqrt15], no root evaluated	✓

Key	Claim	How	Lean
`contraction`	$\lambda^{-4} = 31 - 8\sqrt{15}$	the square of the conjugate unit	✓
`jordan`	$\operatorname{rank} M = 5$, $\operatorname{rank} M^n = 4$ $(n \geq 2)$	exact elimination over Q	
`q_plus_eigen`	$Q_+ M = Q_+$	left eigenvector for the marginal eigenvalue +1	✓
`q_minus_eigen`	$Q_- M = -Q_-$	left eigenvector for the marginal eigenvalue -1	✓
`q_values`	$Q_+ \equiv -1, \quad Q_- = +1, -1, +1, -1$	evaluated on the matrix census at depths one to four	✓
`censuses`	$T = 1, 8, 63, 496$; $T_{\text{geom}} = 1, 9, 71, 559$	the same patches counted with and without the Mystic	
`mystic_rule`	$T_{\text{geom}}(n) = T(n) + T(n-1)$	row Gamma of M is all ones, so N_Gamma(n) is the whole census at n-1 and the Mystic is a second copy of it	✓
`ledger`	$N_\Theta(n) = N_\Gamma(n-1)$	row Theta of M is the indicator of Gamma, so the count is the previous generation of Gamma verbatim	
`ledger_lambda`	$N_\Lambda(n) = N_\Sigma(n-1)$	row Lambda of M is the indicator of Sigma	
`hat_charpoly`	$\chi_H(x) = (x-1)(x+1)(x^2 - 7x + 1)$	characteristic polynomial of the four-metatile Hat substitution	
`hat_perron`	$\kappa = \frac{1}{2}(7 + 3\sqrt{5}) = \varphi^4$	largest root of the Hat quadratic, and the fourth power of the golden ratio	
`phase_fields`	$\mathbb{Q}(\sqrt{15})$ against $\mathbb{Q}(\sqrt{5})$	the two characteristic polynomials differ in a single coefficient, seven against eight, and that...	
`frequencies`	$v \in \mathbb{Z}[\sqrt{15}]^9$, $\sum v = 1$	nullspace of M - lambda^2 I, normalised	✓
`hat_frequencies`	$f_H = \frac{1}{3}$; $f_T, f_P, f_F \notin \mathbb{Q}$	Perron eigenvector of the four-metatile Hat substitution, over Q(sqrt5)	
`aperiodicity_witness`	9/9 vs 3/4 irrational	a periodic tiling has rational frequencies, so an irrational one refutes periodicity	✓
`trace_accounting`	$\text{tr} = \text{unit} + \text{conjugate} :$ 8 , 7	the marginal pair +1 and -1 cancels and the zero modes contribute nothing, so the trace is the unit plus...	✓
`twist_two_levels`	$\chi_B(x) = (x-1)^2(x^2 - 8x + 1)$, $B\,v = \lambda^2 v$	the supertile quadrilateral moves by an exact map, transcribed from the construction	
`spectre_twist`	$\tan\theta_S = \frac{1}{13}(5\sqrt{3} - 6)$, $\cos 2\theta_S = \frac{1}{10}(4 + 3\sqrt{3})$	in the limit of high level, one level relates the quadrilaterals by ratio lambda and rotation minus...	
`twist_field`	$\cos 2\theta_S \in \mathbb{Q}(\sqrt{3})$, $\notin \mathbb{Q}(\sqrt{15})$	the angle lies in the field of the hexagonal tile coordinates	
`boundary_recurrence`	$P(k) = 5P(k-1) - 3P(k-2) - P(k-3)$	proved for every k by the finite certificate boundary_certificate/check_spectre_boundary_proof.py (93...	✓*
`boundary_chi`	$\chi_\partial(x) = (x-1)(x^2 - 4x - 1)$	characteristic polynomial of the companion matrix of the recurrence	
`boundary_growth`	$\nu = 2 + \sqrt{5} = \varphi^3$	Perron root of the boundary operator	
`boundary_field`	$\nu \in \mathbb{Q}(\sqrt{5})$, $\lambda^2 \in \mathbb{Q}(\sqrt{15})$	area and perimeter of the same tile inflate in different quadratic fields	
`boundary_dimension`	$\log\nu / \log\lambda$ (conditional)	the growth exponent of the perimeter against the diameter (the perimeter recurrence is proved for every k)	
`balanced_exponent`	$s^* = 1 - \log\nu / \log\lambda^2$	the thickness exponent making every substitution level carry comparable volume (a modelling choice)	
`compositum`	$\mu = \lambda^2/\nu$ has degree 4	the balanced ratio lies in Q(sqrt3, sqrt5) and in neither quadratic field alone	
`golden_absent`	$(x^2 - 4x - 1) \nmid \chi_M$, χ_H	the boundary quadratic divides neither substitution matrix	
`lucas_deflation`	$\varphi^n :$ $x^2 - L_n x + (-1)^n$	every power of the golden ratio has a Lucas quadratic for its minimal polynomial, so agreement of two such...	
`spectre_not_golden`	$\lambda^2 \neq \varphi^n$	the trace of the Spectre unit is eight, which is not a Lucas number, so the area inflation is not golden...	
`boundary_invariant`	$P(k+2) - 4P(k+1) - P(k) = -16$	the quadratic factor of chi applied to the sequence annihilates the growing part and leaves the constant...	✓*
`boundary_growth_bound`	$P(k+3) \geq 4P(k+2) + 30$	for the sequence defined by the recurrence (proved in Lean) the perimeter more than quadruples at every...	✓*

Key	Claim	How	Lean
`lam_below_four`	$\lambda^2 < 16$	16 - lambdaˆ2 = 12 - sqrt15, positive since 12ˆ2 ¿ 15	✓
`boundary_exact`	$P(k)$ counted exactly in $\mathbb{Q}(\sqrt{3})$	the same sequence recounted with edges identified by equality of exact quadratic-field coordinates rather...	
`chair_mesh`	$(V, E, T) = (2138,\ 6408,\ 4272)$, $\chi = 2$	the 24 panels of the seven-cube chair, each cut by its 9x9 grid into 73 flat rectangles and 8 four-sided...	✓
`chair_volume`	$\mathrm{vol} = 7 + \frac{4}{3} w^2 h \sum_s \gamma_s = 7$ for all w, h	the volume of the symbolic solid, for every base width w and height unit h at once: the signed heights sum...	✓
`chair_labels`	192 sites , $\gamma_s \in \{\pm 1, \ldots, \pm 12\}$, $\sum \gamma_s = 0$	height labels forced by exact complementarity on every shared face of the depth-two patch of the recorded...	✓
`chair_tilings`	$3^8 =$ 6561 rotation tilings of the doubled chair by 8 chairs	exact cover of the 56 cells of the doubled chair by eight rotated copies, by a bitmask search	
`chair_screen`	6555 tilings leave 4 free classes; 6 leave 12, and 3 of those use all 24 orientations	every one of the 6561 tilings is glued to depth two	
`chair_atlas`	2388 raw contacts → 44 legal → 30 realized	every face contact of two chairs (48 frames, translations in a box), the legal ones kept by exact site...	✓
`chair_equations`	372 site equations , rank = 180 , $\dim\ker = 12$	the distinct complement equations c_A + c_B = 0 of the 30 realized contacts, ranked exactly over Q	✓
`chair_chiral`	Sym(decorated) = {1} , while |Sym(chair)| = 6 = 3 + 3	the undecorated seven-cube chair has six symmetries, three of them mirrors, so it is achiral	
`chair_forced_chirality`	no mirror contact is legal: all 44 legal contacts have det = +1	of the 2388 raw face contacts 1194 place a mirror image against the root	
`chair_orientation`	$\chi(x) = x^3(x-8)(x-2)^{20}$ on the 24 rotation species	the orientation transfer matrix of the substitution on the 24 rotations	
`chair_surface`	surface $= 24 \cdot 4^N$, shared $= 12(8^N - 4^N)$, $B_c = 12 \cdot 4^N (2^c - 1)$	counted on the placed patch for N ¡= 4 with the face-directions exactly flat (4ˆ(N+1) per +-x,+-y,+-z)	✓
`chair_exponents`	$D_b = 2$, $s^* = 1/3$, $\mu = 2$ ($\lambda = 2$, $\nu = 4$, $d = 3$)	the unified exponents D_b = log nu / log lambda, s* = 1 - D_b/d, mu = lambdaˆd / nu with the Chair44...	
`chair_twist`	supertile $= 2 \cdot$ chair , twist = 0 , $\frac{1}{8}\sum_i H_i = \frac{1}{4} I$	the 8 child chairs are laid on the doubled lattice and their cells compared with the doubled chair: equal,...	
`chair_nets`	surface = 24 squares , $(V, E, F) = (26, 48, 24)$, 261,434,325,600 spanning trees	the exposed unit faces of the seven-cube chair, their edge-adjacency graph (every edge shared by exactly...	
`chair_slices`	$\sum_j d_j = 2^n(1, 1, 1)$, layer interface $= 4^{n+1} - 4 \cdot 2^n$	the 8ˆn chairs of the depth-n patch, n¡=4, each reduced to the centre g(1,1,1)+t of its bounding cube and...	
`orientation_sums`	$S_n^{\mathrm{chair}} = 2^n$ exactly ; $\lvert S_n^{\mathrm{Spectre}} \rvert / \lambda^n \to 1.4489$	the sum of the orientation vectors over the depth-n patch: for the chair the z component of the notch...	
`power_slopes`	median $I(k)/N$ falls as $N^{-0.34}$ (Spectre) and $N^{-0.72}$ (chair)	the median over 300 fixed random wave vectors (seed 1) in the box [0, 2 pi / l]ˆd, with l the tile size...	
`chair_coincidence`	modular coincidence at depth 3 , $c(M) = 28 \cdot 2^M - 48$ ($M \geq 3$)	the cell substitution on 24 x 7 = 168 labelled unit cells, rebuilt from the child poses: column sum 8,...	✓
`chair_legal_blocks`	1410 legal vertex blocks, closed under σ ; 27 Φ-fixed seeds, stabilisers of order 1 and 8	the 2x2x2 windows of sigmaˆn(j) over the 168 labels, n = 1..6, from the cell substitution of...	
`chair_hull_theorem`	every atlas tiling lies in the hull ; $\mathrm{Sym}(T) \in \{1, C_2, C_4, D_4\}$, $\lvert \mathrm{Sym}(T) \rvert \leq 8$	Romanov (this paper): one level of the hierarchy plus a finite computation on the 8 cells around a lattice...	
`chair_lines`	box means of the height and notch observables are 0 ; 6 chair stacks give diagonal lines with means $\mp 36, \pm 1$	exact, from the cell substitution of chair44_twoadic	
`boundary_fibonacci`	$P(k) = 2F_{3k+5} + 4$	for the sequence defined by the perimeter recurrence (seeds 14, 46, 182, characteristic polynomial...	✓*
`boundary_directions`	even $= F_{3k+4} + 1$, odd $= F_{3k+3} + 1$ ($k \leq 6$)	boundary edges of an order-k supertile, sorted by direction mod 180 degrees (the 14 edges point in...	✓*
`boundary_z3_bounded`	$\max \lvert \hat{h}(j) \rvert$ over the Z/3 and faithful modes $\leq \sqrt{13}$ ($k \leq 6$)	the DFT of the folded direction histogram over Z/6 = Z/2 x Z/3 (the split spectre_torsion uses for the six...	

Key	Claim	How	Lean
`centre_diffraction`	$X_n = 2X_{n-1} \cup (\text{odd box} \setminus \text{corner octant})$; $\sum_k I(k) = \frac{1}{7} = \text{density}$	the centre of every chair of the depth-n patch obeys the nested recursion above (checked against the...	
`orientation_recursion`	$S_{n+1} = C\,\overline{S_n}$, $\lvert S_n \rvert \sim c\,\lambda^n$, $c^2 = \frac{15+5\sqrt{3}+3\sqrt{5}+3\sqrt{15}}{20}$	the sum of exp(i angle) over the tiles of the depth-n supertile of each label obeys a linear recursion...	
`orientation_channels`	charpoly $= x^3(x-8)(x-2)^{20}$ (chair) ; $h_m = 2, 1, 0$ (Spectre)	the 24 orientation states of the Chair44 are split by the irreducible representations of the rotation...	
`contact_matrices`	Spectre: 14×14, rank 14, bipartite ; Chair44: 24×24, rank 24, not bipartite	the number of times native edge a of one tile meets native edge b of a neighbour, counted over the level-4...	
`lift_integer`	every vertex has integer coordinates $(p, q, r, s) \in \mathbb{Z}^4 = \mathbb{Z}[\zeta_{12}]$; rank $= 4$	every edge of the patch is a unit vector at a multiple of 30 degrees: even multiples are p+qw with w=exp(i...	
`inflation_matrix`	charpoly $= (x-1)^4(x+1)^4(x^4-8x^2+1)^2$, $\lambda^2 = 4 + \sqrt{15}$	the supertile recursion of spectre_patch moves four key points by rotations through multiples of 60...	
`internal_plane`	bounded plane over $\mathbb{Q}(\sqrt{15})$; $\dim_\partial$ of the window ≈ 1.2	a linear functional is bounded on the supertile corners exactly when it kills the expanding part...	